

2. Requirement Analysis

Project Name

ComicCraft - AI Comic Story Creator using Gemini Models

1. Introduction

ComicCraft is an AI-powered comic story creation application that helps users create creative comic stories using Google's Gemini AI models. The system generates story content based on the user's input and helps organize the story into comic-style scenes.

2. Problem Statement

Creating a complete comic story manually requires creativity, time, and writing skills. Beginners may find it difficult to develop characters, dialogues, scenes, and a proper storyline.

ComicCraft aims to simplify this process by using Generative AI to help users create comic stories quickly and easily.

3. Objectives

- To create comic stories using AI.
- To generate creative story ideas from user input.
- To generate characters, scenes, and dialogues.
- To provide an easy-to-use interface.
- To reduce the time required to create a comic story.
- To use Gemini AI for intelligent story generation.

4. Functional Requirements

The system should allow the user to:

1. Enter a story idea or topic.
2. Select or provide character details.
3. Generate a comic story using Gemini AI.
4. Generate scenes and dialogues.
5. Preview the generated comic content.
6. Edit or regenerate the generated content.
7. View the final comic story.
8. Non-Functional Requirements

- The application should be easy to use.
- The system should provide responses within a reasonable time.
- The interface should be simple and user-friendly.
- The application should handle user input properly.
- The generated content should be relevant to the user's prompt.
- The system should maintain reliable communication with the AI service.

6. Hardware Requirements

- Computer/Laptop
- Minimum 4 GB RAM
- Internet connection
- Keyboard and mouse

7. Software Requirements

- Operating System: Windows/Linux
- Frontend: HTML, CSS, JavaScript
- Backend: Python Flask
- AI Model: Google Gemini
- Code Editor: Visual Studio Code
- Browser: Google Chrome
- Version Control: Git and GitHub

8. User Requirements

The user should be able to:

- Enter a simple story idea.
- Provide character information.
- Generate a story with AI.
- View the generated scenes and dialogues.
- Preview the completed comic story.

9. System Requirements

The system requires:

- A web browser.
- Internet connectivity.
- Python environment for backend execution.
- Gemini API access for AI-based story generation.

10. Expected Outcome

The expected outcome of ComicCraft is a simple web application that uses Gemini AI to transform a user's story idea into a creative comic story with characters, scenes, and dialogues.

11. Conclusion

The requirement analysis defines the functional, non-functional, hardware, and software requirements of the ComicCraft project. These requirements provide a foundation for designing and developing the AI-powered comic story creation system.